

School Information System: Comprehensive Update Report

This report outlines the extensive refactoring, bug fixing, and design enhancements we have implemented across the Galala SIS platform to transform it into a secure, professional, and fully decoupled academic management system.

1. System-Wide Architectural & Security Upgrades

- **Role-Based Access Control (RBAC):** We implemented strict, iron-clad access boundaries across the system.
 - **Administrators** are now strictly focused on administration. They can no longer access, create, or modify academic assignments or take attendance. All academic routes (`/assignments`, `/attendance/save`) now return a 403 Forbidden error if an admin attempts to access them.
 - **Teachers** can only view and manage data that belongs to them. We added deep backend validation to ensure a teacher cannot manipulate URLs (e.g., `?subject_id=999`) to view or grade students in subjects they do not teach.
 - **Students** are strictly isolated. Their dashboards and assignment views cross-reference the `Enrollments` table so they only see assignments and data for subjects they are actively enrolled in.
- **AI Assistant Relocation:** The AI Assistant module was removed from the Student and Teacher views and made an exclusive tool for the Administrator.

2. Decoupling Assignments and Final Grades

- **The Overwrite Bug:** We identified and eradicated a critical legacy bug where grading a single assignment would automatically overwrite and wipe out a student's official `Final_Grade` in the `Enrollments` database.
- **New Gradebook Module:** We developed a dedicated, professional Gradebook for teachers. Teachers can now select their assigned subjects, view a roster of enrolled students, and input their official Final Semester Grades completely independent of daily assignment scores.
- **Student Profile Separation:** We redesigned the backend queries so that a student's profile cleanly separates their "Final Enrolled Grades" from their daily "Assignment Submissions."

3. UI/UX "Ultra-Premium" Redesigns

We overhauled the frontend interfaces, moving away from basic custom CSS to a modern, glassmorphic design system utilizing advanced Bootstrap 5 features.

- **Dashboards:** Upgraded the Admin, Teacher, and Student dashboards with vibrant gradient hero banners, floating metric cards with soft shadows, and clean, responsive layouts.
- **My Grades & Student Profiles:** Completely rebuilt `student_profile.html` and `my_subjects.html`. Replaced loud, generic colors with a clean dashboard-

surface white aesthetic, utilizing crisp typography, subtle colored icon accents, and rounded pill badges for grades and statuses.

- **Date Input Restrictions:** We configured all date inputs across the system to enforce logical constraints (e.g., Date of Birth cannot be set to a future date).

4. Dashboard Data Accuracy & Scoping

- **Teacher Dashboard Fixes:** Fixed the "Recent Student Enrollments", "Total Students", and "Top Students" stat cards. They previously showed school-wide data; they now accurately calculate data exclusively from the specific students enrolled in the logged-in teacher's subjects.
- **Student Dashboard Fixes:** Fixed the "Available Assignments" counter and "Upcoming Assignments" widget. They now execute complex joins against the `Enrollments` table to ensure a new student with zero enrollments sees zero assignments, rather than the total assignments across the entire school.

5. Automated System Testing

- **Database Seeding (`seed_db.py`):** We engineered a comprehensive backend script that populated the system with dozens of fake records (Departments, Classrooms, Subjects, Teachers, Students, Enrollments, Schedules, and Assignments).
- This allowed us to simulate a live, heavily-populated school environment to aggressively test the new role-based security barriers and ensure the dashboard queries functioned perfectly under scale.